

**CareLink: A Smart Health Referral System with Automated Queue Prioritization
for Barangay and Koronadal City Health Centers**

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Approval Sheet

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Abstract

Acknowledgments

Table of Contents

Approval Sheet	2
Abstract	3
Acknowledgements	4
List of Tables	7
List of Figures	8
List of Appendices	9
Introduction	10
Background of the Study	10
Objectives of the Study	10
Review of Related Literature	10
Brunauer-Emmett-Teller (BET) surface area.....	10
Topic 2.....	10
Topic 3.....	10
Related Studies	11
RRS1.....	11
RRS2.....	11
RRS3.....	11
Synthesis.....	11

Significance of the Study	12
Scope and Limitations	12
Operational Definition of Terms	12
Method	14
System Methodology	14
Requirements Analysis	14
Data Flow Diagram of the Current System.....	14
Use Case Diagram of the Current System.....	14
Requirements Documentation	15
System Requirement for Hardware.....	15
Software.....	15
User Requirements.....	15
Use Case Diagram of the Proposed System.....	15
Activity Diagram.....	15
Modules of the System.....	15
Design of the System	15
Conceptual Design.....	15
Entity Relationship Diagram.....	15
Class Diagram.....	16
Data Schema of the System.....	16

Data Dictionaries.....	16
Results and Discussions	17
Development and Testing	17
System Usability Scale (SUS).....	17
Description of the System	17
Implementation Plan	17
Recommendations	18
References	19
APPENDICES	20

List of Tables

Table 1. Input Parameters	23
Table 2. Results of...	24
Table 3. Comparison of	24
Table 4. Physical Properties of the Sand	25

List of Figures

Figure 1. Literature Map	10
Figure 2. Conceptual Framework of the Study	11

List of Appendices

Appendix A Screenshots of the Testing of the Prototype	30
Appendix B Official Testing Results (optional)	31
Appendix C Evaluation Responses (optional)	32
Appendix D Pictures showcasing the actual design/development of the prototype	33
Appendix E Forms	34

Introduction

Background of the Study

Health systems in the Philippines are primarily dependent upon the services rendered by barangay health stations and city health offices in attending to the healthcare needs of their citizens. As stated in Republic Act No. 11223 or the "Universal Health Care Act of 2019," the local government units must provide their constituents with timely and quality health services. Nonetheless, the provision of health services continues to be problematic due to inefficiency in their health care service delivery. The PCHRD (2022) mentioned that the bulk of public health facilities still depend on the manual way of processing patients and referrals from one facility to another.

Koronadal City in South Cotabato is run by barangay health centers and Koronadal City Health Centers on purely manual systems. Patients form a physical line without any way of keeping track of their place in the line or even how much longer they will be queuing. Patients receive referral slips manually, which they carry themselves. They join the walk-in line again after arriving at the Koronadal City Health Center, irrespective of the level of urgency of their conditions. Referral slips are commonly lost during transit, with no way to verify referral completion. On average, referral completion rates in the Philippines range from 52 to 65% (PCHRD, 2022).

Evidence from comparable health systems demonstrates the measurable impact of digitizing referral workflows. The STAHR e-referral initiative at Rizal Medical Center reduced average referral processing from over 24 hours to under 3 hours (Novartis Philippines, 2024), while a study at East Avenue Medical Center reported a 63%

reduction in referral turnaround a time after implementing a structured digital referral system (Tan, 2025). (Adhicandra et al., 2024) further demonstrated that a rules-based priority queue algorithm significantly reduced emergency patient waiting times and improved service fairness, and (Presume et al., 2024) confirmed that automated SMS reminders substantially improved appointment attendance in resource-limited health facilities.

To address the problems that have been raised in this research, this study recommends the introduction of CareLink: A Smart Health Referral System with Automated Queue Prioritization at Barangay and Koronadal City Health Centers. This is a web-based system that seeks to automate the referral process, assign automated queue priority based on a Weighted Sum Model priority scoring algorithm ordered through a priority queue, monitor referral progress based on a multi-state workflow, and send automated SMS notifications to patients and health workers. Through the use of the functionalities described above in one comprehensive platform appropriate to the needs of the healthcare facilities of Koronadal City, improvements in referrals will be realized.

Objectives of the Study

The main objective of this study is to design, develop, and evaluate CareLink: A Smart Health Referral System with Automated Queue Prioritization for Barangay and Koronadal City Health Centers. The proposed system aims to improve referral coordination, reduce patient waiting time, enhance communication between health facilities, and support efficient healthcare service delivery through digital technologies, automated workflows, and intelligent queue prioritization.

Specifically, it aims to:

1. Design and develop a web-based digital referral management system with patient registration, digital referral form submission, and automated SMS notifications for referral approval, appointment reminders, and missed referrals.
2. Implement a referral tracking module that monitors referral status in real time, from submission to completion, between barangay and city health centers.
3. Implement a Queue Prioritization Algorithm, using a Weighted Sum Model and a priority queue, that classifies referred patients into Priority 1 (Emergency), Priority 2 (Vulnerable Group), and Priority 3 (Standard Referral) based on clinical urgency, demographic classification, and referral type.
4. Develop advanced analytics and reporting features covering peak hour analysis, average service time, staff performance, queue abandonment rate, patient satisfaction trends, and queue volume forecasting.
5. Evaluate the developed system in terms of:
 - a. System Usability using the System Usability Scale (SUS);

- b. User Satisfaction using a Technology Acceptance Model (TAM)-aligned survey measuring perceived usefulness and perceived ease of use; and
- c. Performance Efficiency using response time metrics for referral submission, queue prioritization, referral processing, and SMS notification trigger latency.

Review of Related Literature

This section reviews related literature and systems that will help researchers gather basic information and references for the current study.

Health Information Systems in Primary Care

Health information systems are technological tools developed to assist in managing healthcare data, aid decision-making and increase efficiency in the provision of health services at all levels. The importance of HIS has greatly increased in primary care settings due to the high number of patients served, making effective management of their data very important. Resource-limited environments have been shown to require moving from manual to electronic processing of information as a fundamental requirement for improving health services provided.

In their review of the use of health information systems in developing countries, (Bagherian & Sattari, 2022) discovered that workflows that utilize paper documents always lead to fragmentation of data, duplicated services, and delayed clinical decisions. The authors stressed that the factors for successfully implementing an HIS within a setting where technology may be lacking include ease of use, low bandwidth usage, and

compatibility with existing facility workflows. This is precisely the case with the system used in CareLink.

In the national scene, the PCHRD (2022) cited the total lack of digital referral networks in Philippine barangay and city health centers as one of the biggest gaps in the Philippines' primary care network, which made it necessary to have an innovative healthcare solution like CareLink. In addition, (Garcia et al., 2022), in their cross-sectional study published in the Journal of Medical Internet Research, looked at health information systems within selected government tertiary hospitals and health centers in the Philippines and revealed fragmentation in data management as well as digital information infrastructure to still be very common within Philippine government health facilities.

The security and integrity of health information systems further depend on clearly defined role-based access structures that restrict data access according to each user's responsibilities within the facility. (Fareed & Yassin, 2022), publishing in the Bulletin of Electrical Engineering and Informatics, proposed a role-based access control scheme for e-healthcare systems and demonstrated that assigning differentiated access permissions based on user roles from administrators to data entry staff significantly improved system security, reduced unauthorized access risks, and supported the enforcement of organizational data policies without disrupting routine workflows. These findings directly inform CareLink's role-based architecture, which defines four distinct user roles Super Admin, Barangay Health Center Staff, City Health Center Staff, and Patient each with access permissions scoped specifically to their operational responsibilities within the referral and queue management workflow.

Patient Referral Systems in Healthcare

Referral systems for patients are processes where patients are referred from one level of the health care system to another in order to be provided with health care services not available in the referring health care facility. Referral systems are an integral part of any primary care system as they allow for continuity of care and help avoid disruptions in treatment for patients who are unable to receive treatment in the referring health care facility.

The World Health Organization (2022) identifies a well-functioning referral system as one that ensures the right patient receives the right care at the right time. In the Philippine public health system where barangay health centers serve as the primary entry point, efficient referral coordination with city health facilities is a matter of direct patient safety. (Montellier et al., 2022), in a study published in BMC Primary Care, demonstrated that an electronic referral system designed for direct hospital admissions showed strong potential to facilitate faster inter-facility coordination compared to manual referral processes, with results confirming the system's capacity to facilitate admissions within shorter time frames.

More specifically, regarding the case of the Philippines, (Aliguin & Baun, 2025) in their descriptive research study investigating the use of digital communications in referral processes of inter-hospital systems in Pangasinan have concluded that most of the health institutions still depend on non-formal channels like Facebook Messenger to coordinate their referrals, while at the same time pointing out the impact of technical and personal issues on their effectiveness. Therefore, CareLink meets the requirements stated by the

researchers regarding the need to create a specialized digital referral system. Such needs are met using the Digital Referral Form Module as well as the Multi-State Referral Tracking Workflow.

Queue Management and Priority Algorithms in Health Facilities

Queue management systems in healthcare are designed to organize patient flow, reduce waiting times, and ensure that urgent cases are given appropriate priority. In public health centers that serve high patient volumes with limited staff, unmanaged queues lead to prolonged waiting times, patient dissatisfaction, and inequitable service delivery where vulnerable patients are treated the same as standard walk-in cases. The integration of algorithmic priority classification into queue management has emerged as a practical solution to these challenges.

(Adhicandra et al., 2024), publishing in the International Journal of Software Engineering and Computer Science, demonstrated that a priority queue algorithm combined with reinforcement learning significantly reduced emergency patient waiting times and improved service fairness across patient types. CareLink adapts only the deterministic priority queue component of this approach, consistent with the study's scope, which excludes machine learning-based prioritization. Their findings support CareLink's Intelligent Queue Priority Assignment Module, which uses a Weighted Sum Model to compute each patient's priority score from clinical urgency, demographic classification, and referral type, then orders patients in a priority queue by descending score, with timestamp used as a First-Come-First-Served tiebreaker, to classify them into

Priority 1 (Emergency), Priority 2 (Vulnerable Group), and Priority 3 (Standard Referral) without manual staff intervention.

Complementing this, (Mohd Wafi Nasrudin et al., 2023) found that a smart waiting system for outpatient clinics using mobile queue numbering eliminated the need for patients to remain physically present while waiting, improving patient satisfaction and reducing unnecessary congestion at health facilities. This finding directly supports the value of CareLink's Queue Status Viewing Module, which allows referred patients to monitor their queue position remotely rather than waiting on-site. Together, these studies validate CareLink's approach of combining automated priority assignment with real-time queue visibility as a means of improving both operational efficiency and patient experience at the city health center level.

Electronic Health Records in Philippine Primary Care

It is evident that the use of paper documents as compared to electronic health documents in Philippine communities has been found to offer significant benefits in terms of efficiency at all levels. Paper documents are usually lost easily, difficult to read, and can be duplicated, thus making it difficult to ensure continuity in health care services especially when the patients go from one barangay health center to another in the city.

The study carried out by (Elepaño et al., 2025), which involved the adoption of an electronic health record system at the Barangay health station under the University of the Philippines Health Service through a pilot study using a mixed-methods approach, showed that the use of e-health records led to more accurate and complete patient records, fewer duplicates, and allowed health workers to have access to patient data in

subsequent consultations. The research was able to identify specific barriers to implementation in the Philippine barangay setting, such as unreliable internet connections and easy-to-use interfaces.

Similarly, (Elepaño et al., 2025) in their mixed-methods study on implementation published in JMIR Medical Informatics demonstrated the validity of the above claims using two EHR systems implemented in both urban and rural healthcare facilities under the Philippine Primary Care Studies Program. The study demonstrated that acceptability of EHR system use was high in settings where the user interface was simple and aligned with workflow processes at the health facility. Thus, in light of the above studies, one can confidently say that the design philosophy used in designing CareLink's patient record management tool in terms of a simple user interface and offline capabilities has been validated in the Philippine context.

SMS Notification and Automation in Health Services

Short Messaging Services (SMS) still remain one of the most available and dependable means of communication in the Philippines, especially in places where there is poor internet access. As opposed to smartphone applications and email, SMS does not necessarily require an internet connection and a specific application, which makes it an ideal tool of communication in terms of reaching out to patients regardless of their social background. This can be very important when it comes to reducing patient dropouts during health referrals.

According to (Presume et al., 2024), a research study conducted among four different healthcare institutions in Haiti has established that patients in clinics with SMS

automated reminders were significantly more likely to attend their appointments than those in clinics without reminders. The results of this study validate the use of SMS automation technology as a cost-effective method that works well in environments that are similar to the Philippines' community-based clinics.

(Alturbag, 2024), in a study published in Cureus, further reinforced these findings by evaluating personalized SMS and phone call reminders among 493 patients at a radiology department in Saudi Arabia, finding that reminder interventions produced a measurable improvement in overall patient attendance rates across diverse healthcare settings. Together, these studies validate CareLink's SMS Notification and Automation Module, which implements a three-trigger workflow a referral approval alert with assigned queue number, a 24-hour appointment reminder, and a missed referral alert each designed to address a specific point of patient dropout in the referral pathway.

Technology Acceptance in Health Information Systems

The success of implementation of health information systems is not only determined based on their functionality from a technical perspective, but by whether or not the end-users, especially healthcare professionals, see these systems as effective and user-friendly. Even if the system itself works flawlessly, it will fail in application due to its poor uptake by intended users. For this reason, the study of factors influencing adoption of technologies in healthcare is necessary.

In a systematic review of 29 articles on the application of the Technology Acceptance Model in health information systems conducted from 2018 to 2023 by (Tetik et al., 2024), it was established that there are two main factors that contribute to health IS

adoption – perception of usefulness and perceived ease of use. The results presented by (Tetik et al., 2024) serve as a theoretical basis for applying TAM to CareLink, wherein health practitioners from different levels, including barangay level health centers, will be working within the system in various conditions of digital competence and pressure. Thus, measurement of these two factors is crucial in determining the adoption of CareLink in an LGU health context.

(Al-Momani & Ramayah, 2023), in a review published in the International Journal of Business and Technology Management, further confirmed that TAM has been consistently adapted and extended across healthcare settings to capture context-specific factors influencing technology adoption, and that perceived usefulness and perceived ease of use remain the most reliable predictors of health IS acceptance. These findings directly inform CareLink's evaluation approach, which combines a TAM-aligned user satisfaction survey with the System Usability Scale (SUS) to measure both usability and sustained adoption intent among LGU health workers. Together, these instruments provide a comprehensive assessment of whether CareLink meets the practical and experiential needs of its intended users.

Analytics and Reporting in Health Information Systems

Data analytics and reporting functionalities have become essential components of modern health information systems, enabling administrators and health workers to monitor service delivery trends, evaluate system performance, and support evidence-based decision-making at the facility level. In primary care settings where referral volumes and queue patterns fluctuate across time and population groups, visual

reporting tools provide the operational visibility needed to allocate resources equitably and identify service bottlenecks before they affect patient outcomes.

(Xie et al., 2022) carried out a review of randomized control trials in the journal of the American Medical Informatics Association, investigating the effectiveness of the use of dashboards as an audit and feedback tool in both primary care and hospital contexts. (Xie et al., 2022) review demonstrated that using a dashboard as a reporting tool facilitated the monitoring of clinical processes and ensured that administrators were provided with easily understandable summaries of key performance indicators that needed to be manually computed using traditional means. The above findings clearly justify the inclusion of CareLink Analytics and Reporting Dashboard that provides City Health Center Staff and administrators with visual summaries of various performance metrics including number of referrals, queue performance, distribution priorities, completion rates, and SMS notification results.

(Epizitone et al., 2022), in a bibliometric analysis of health information system performance published in Healthcare (MDPI), confirmed that analytics and reporting capabilities have become among the most consistently valued features of HIS implementations globally, as they enable health administrators to shift from reactive to proactive service management by making system-generated data visible and actionable. Their findings reinforce CareLink's design decision to embed reporting and analytics as a core module rather than an optional feature, ensuring that administrators at the LGU level have continuous access to referral and queue data without requiring external tools or manual report generation.

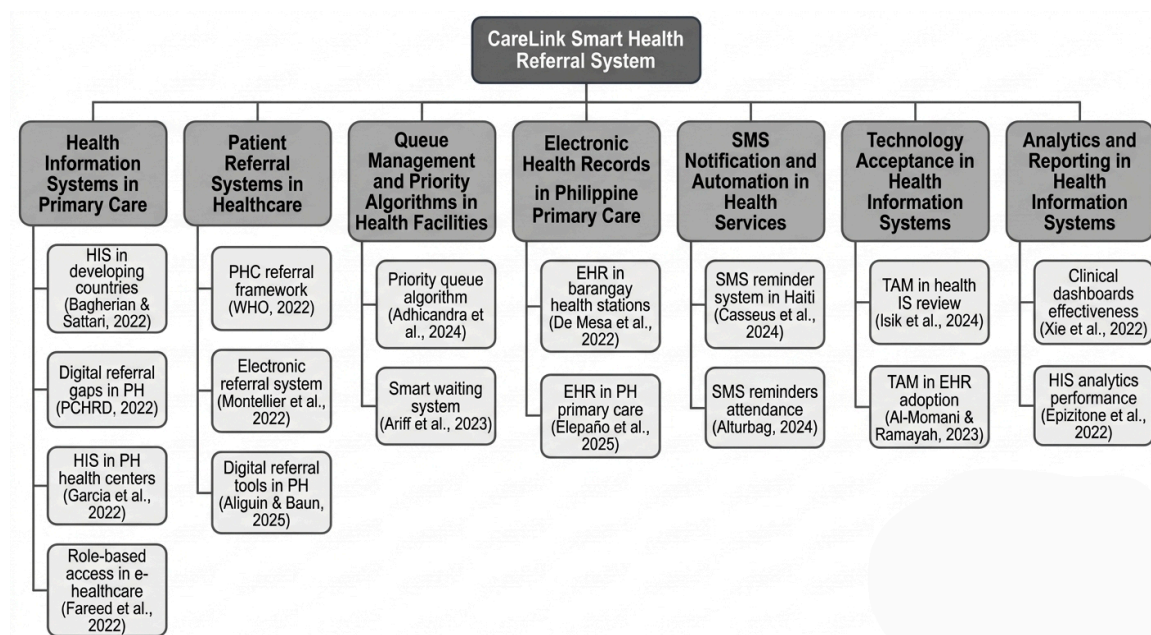
Figure 1.*Literature Map*

Figure 1 presents the literature map of CareLink, organizing the academic and empirical foundations of the study into seven thematic clusters: health information systems in primary care, patient referral systems in healthcare, queue management and priority algorithms in health facilities, electronic health records in Philippine primary care, SMS notification and automation in health services, technology acceptance in health information systems, and analytics and reporting in health information systems. Each theme is supported by peer-reviewed and published sources that collectively establish the theoretical grounding, contextual relevance, and technical basis for the design and development of CareLink.

Related Studies

This section presents studies conducted to which the present proposed study is related or has some bearing or similarity.

Introducing a Digital Patient Referral System to Philippine Healthcare

The Philippine Council for Health Research and Development (PCHRD, 2022) assessed the current state of referral systems in Philippine public health facilities and proposed a framework for digital referral platform adoption. The study found that most public health facilities, including barangay health centers and city health offices, continue to rely on manual referral processes marked by lost referral documents, absent feedback mechanisms, and inequitable treatment of referred patients who must rejoin queues from the beginning. The study recommended the development of web-based referral platforms integrating patient data management, real-time referral tracking, and inter-facility communication. CareLink directly responds to every gap identified in this study by providing exactly these features, contextualizing them for the specific operational environment of Koronadal City's barangay and city health centers.

Improving Emergency Referrals in a Philippine Tertiary Hospital

(Tan, 2025) conducted a quality improvement study at East Avenue Medical Center focused on reducing inefficiencies in the emergency referral process. The study found that manual referral workflows produced turnaround times exceeding 43 minutes and documentation compliance of only 75%. After implementing a structured digital referral system, referral processing time decreased by 63% and documentation compliance improved to 98%. While focused on a tertiary hospital context, the magnitude of improvement achieved through digitizing and standardizing the referral process provides strong empirical support for CareLink's Digital Referral Form Module and Referral Status Tracking Module, both of which apply the same principle of replacing unstructured manual processes with a standardized digital workflow.

Implementing Electronic Health Records in Philippine Primary Care Settings

The implementation of EHR systems was studied through a pilot mixed-method study by (Elepaño et al., 2025), in JMIR Medical Informatics, involving barangay health stations under the University of the Philippines Health Service. It was found that the use of digital records had positive effects such as increased accuracy of recordkeeping, avoidance of duplicate consultations, and easy access to medical histories. Nevertheless, several difficulties were noted, such as the lack of consistent internet connections and the necessity for streamlined interfaces to fit the skills of varying levels of proficiency of health workers. This study influenced the design process of CareLink through the development of its Patient Registration module and Patient Records module using a light web interface and easy navigation without the requirement of any technical knowledge.

STahr E-Referral Initiative at Rizal Medical Center

The STahr (Strengthening Technology-Use for Improved Access to Health and Referral) project was a collaborative effort between Rizal Medical Center, Thinking Machines Data Science, and Novartis Philippines used to implement the concept of the Health Care Provider Network model of the DOH. After its implementation in September 2023, STahr was able to attain an 88% improvement in referral response time by shortening the average time needed to process referrals from more than 24 hours to less than 3 hours. Even though the STahr intervention program is focused on improving referrals in tertiary hospitals within a vast health network, the significant improvements gained through this system provide direct evidence for the necessity of designing CareLink. Unlike STahr, which focuses on improving specialist referrals in apex

hospitals, CareLink will deal with the barangay-city health centers referral system, a service scope not covered by STAHR.

Smart Management Waiting System for Outpatient Clinics

(Mohd Wafi Nasrudin et al., 2023), from Universiti Malaysia Perlis, developed and evaluated a smart queue management system for outpatient clinics using mobile application-based queue numbering. The system allowed patients to obtain queue numbers and monitor their position remotely without needing to remain physically present at the facility, resulting in reduced congestion and improved patient satisfaction. The study is directly relevant to CareLink's Queue Status Viewing Module and Queue Priority Assignment Module. While the Malaysian system addressed general walk-in queue management, CareLink extends this concept by integrating queue management with the referral approval workflow ensuring that referred patients are automatically assigned a priority queue position upon referral approval, without requiring manual queue insertion by staff.

Implementation of a Patient Reminder System in Haiti

The reliability of an automated appointment reminder system implemented at four healthcare institutions in Haiti was assessed between January 2021 and November 2023 by (Presume et al., 2024). The authors discovered that the appointment turnout rate increased due to the usage of automated SMS and email notifications. The reminder system functioned well despite socio-political instability and infrastructural problems. The results of this study are important for CareLink because they illustrate the effectiveness of an automated system for sending patient notifications under similar conditions of limited access to connectivity. The SMS three-reminder system used in

CareLink (notification about approval, appointment reminder, and missed referral reminder) is based on the approach tested by the authors.

Optimization of Hospital Queue Management Using Priority Queue Algorithm

(Adhicandra et al., 2024) published a study in the International Journal of Software Engineering and Computer Science demonstrating that a rules-based priority queue algorithm integrated with reinforcement learning significantly improved emergency patient service times and queue fairness in a hospital setting. Its foundational finding that priority-based queue classification improves waiting time equity is directly applicable to CareLink's Intelligent Queue Priority Assignment Module. CareLink implements only the deterministic component through a Weighted Sum Model, computing each patient's priority score from clinical urgency, demographic classification, and referral type, then ordering patients in a priority queue by descending score with a First-Come-First-Served tiebreak. This avoids the added complexity of machine learning, making it feasible within the four-month timeline and appropriate for LGU deployment.

Synthesis

This literature review provides the necessary theoretical and empirical foundation, as well as context for CareLink development, based on the relevant literature and studies considered in this chapter. The literature on health information systems showed that paper-based processes in developing countries' primary care settings result in fragmented information, poor coordination, and delay in delivery of services exactly as it was observed in the barangay health centers in Koronadal City (Bagherian & Sattari, 2022); PCHRD, 2022). The literature on patient referral systems suggested that electronic and

standardized referral process results in shorter processing time and improved compliance with documentation requirements, with local research on this issue being available for the East Avenue Medical Center (Tan, 2025) and the STAHR project (Novartis Philippines, 2024). The literature on queue management and priority algorithms demonstrated that rules-based classification of queues results in more equal waiting times and faster access to urgent cases (Mohd Wafi Nasrudin et al., 2023); (Adhicandra et al., 2024). The (Elepaño et al., 2025) pilot study on EHR implementation in Philippine barangay health stations provided specific design guidance on interface simplicity and low-bandwidth optimization for the CareLink patient records component. The literature on SMS automation (Presume et al., 2024) and technology acceptance (Tetik et al., 2024) further validated the inclusion of automated SMS notification logic and TAM-aligned evaluation as integral components of the system.

The studies mentioned above served to prove that each of the components of CareLink has been proven effective within other health systems. However, the lack of an important aspect becomes obvious in all studies as well as within all reviewed health information systems. None of the mentioned studies and the implemented HIs has managed to integrate all the aspects of CareLink – from a rule-based scoring algorithm for prioritization to an analytics dashboard – within one system specifically designed for the process of referral of patients from barangay health centers to those in the city. The STAHR project works on the tertiary care level; the EHR implementation in the study by (Elepaño et al., 2025) does not imply the processes of referral; (Mallari et al., 2022) implements queue management without the processes of referral, etc. Thus, CareLink will not be a copy of the systems under analysis.

Concept of the Study

Figure 2.

Conceptual Framework of the Study

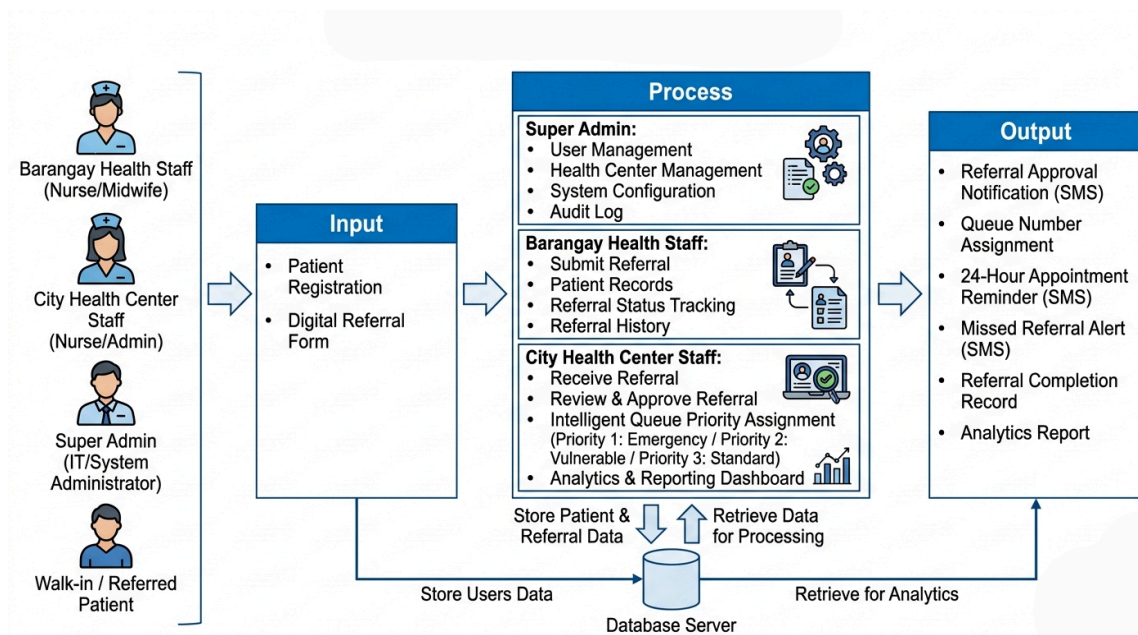


Figure 2 shows the conceptual framework of the CareLink project, depicting its flow from input to process and ultimately into the output of the system. There are four different users of the CareLink System who use the system namely: Barangay Health Staff, City Health Center Staff, Super Admin, and Walk-ins/Referred Patients. Inputs include patient registration and the submission of the digital referral form. These are processed by each user depending on the respective duties performed by each. Super Admin does user management, system administration, and log auditing of activities in the system, Barangay Health Staff submits referrals, manages patients, and monitors their referral status while City Health Center Staff approves referrals and assigns intelligent queuing (emergency, vulnerable or standard). All these processes are recorded in the system through database servers. Outputs include: SMS alerts for approval, queue

numbers, 24-hour reminder calls, missed referrals, referrals completed and analytics report.

Significance of the Study

This study on the development of CareLink: A Smart Health Referral System with Automated Queue Prioritization for Barangay and Koronadal City Health Centers holds significant value for the following beneficiaries:

For Patients. Directly, CareLink assists patients since they will not have to queue for a long time as well as those who have been referred, who will no longer have to start queuing at the very beginning after being referred to the city health center. With the aid of SMS alerts, patients will be notified about the approval of the referral, queue number assigned, and appointments made. Patients who are vulnerable, elderly, pregnant mothers, and PWDs will be the greatest beneficiaries of the system due to the priority status granted to them.

For Barangay Health Center Staff. The system streamlines the referral submission process for barangay health workers by replacing handwritten referral slips with a standardized digital referral form. Staff can monitor the status of submitted referrals in real time and maintain accurate digital patient records, reducing administrative burden and eliminating the risk of lost or damaged referral documents.

For Koronadal City Health Center Staff. The health center team benefits from a streamlined system for handling referrals since the process is more structured than haphazard. The intelligent priority assignment module makes sure that there is no need for any manual intervention because patients are assigned priority automatically through

this module. This way, the health center personnel can pay their full attention to the actual work without having to worry about other things.

For the Local Government Unit of Koronadal City. CareLink provides the LGU with a purpose-built digital health coordination tool that supports compliance with the Universal Health Care Act of 2019 (Republic Act No. 11223), which mandates LGUs to deliver accessible and quality health services. Improved referral completion rates and documented health data support evidence-based planning and resource allocation at the city level.

For The Department of Health (DOH) and Health Policymakers. The system contributes a replicable model for digitizing barangay-to-city health center referral pathways in Philippine LGU settings. Findings from the study's evaluation may inform broader policy on digital health infrastructure development at the primary care level.

For Future Researchers. This study serves as a reference for future researchers exploring health information systems, digital referral platforms, queue management algorithms, and SMS automation in resource-limited Philippine public health contexts. The system architecture, evaluation methodology, and findings contribute to the growing body of literature on primary care digitization in developing countries.

Sustainable Development Goals. This study also aligns with the United Nations Sustainable Development Goals (SDGs), particularly SDG 3: Good Health and Well-Being, which seeks to ensure healthy lives and promote well-being for all at all ages, and SDG 10: Reduced Inequalities, which aims to ensure equal access to essential services regardless of social or economic status. By digitalizing the referral process,

automating queue prioritization for vulnerable groups, and improving coordination between barangay and city health centers, CareLink contributes to the attainment of SDG 3.8 achieving universal health coverage and access to quality essential health-care services while addressing health service inequalities experienced by senior citizens, pregnant women, and persons with disabilities (PWDs) at the community level in Koronadal City, South Cotabato.

Scope and Limitations

This study covers the design, development, and evaluation of CareLink: A Smart Health Referral System with Automated Queue Prioritization for Barangay and Koronadal City Health Centers. The system is intended for use by three types of users: the Super Admin, Barangay Health Center Staff, and Koronadal City Health Center Staff, as well as a patient-facing interface for walk-in and referred patients.

The scope of the system includes a digital referral form module, a barangay-level severity classification module, referral status tracking, patient registration and records management, an intelligent queue priority assignment module using a Weighted Sum Model and priority queue, automated SMS notifications (referral approval, appointment reminder, and missed referral alert), and an analytics dashboard with advanced features (peak hour analysis, service time, staff performance, queue abandonment rate, satisfaction trends, and volume forecasting). The system is web-based and designed for selected barangay health centers and the Koronadal City Health Center.

The evaluation of CareLink covers three dimensions: system usability using the System Usability Scale (SUS), user satisfaction using a TAM-aligned post-use survey

measuring perceived usefulness and perceived ease of use, and performance efficiency using response time metrics for referral submission, priority score computation, queue assignment, and SMS trigger latency.

The study scope was limited to the referral chain from barangay health centers to the Koronadal City Health Center only, with no provisions for referring patients to tertiary facilities, other government institutions, or private healthcare facilities. It does not use machine learning; it uses a deterministic Weighted Sum Model with predetermined weights, ordered through a priority queue. CareLink is not an EMR system and lacks clinical decision support, disease diagnosis, and prescription management tools. SMS reminders depend on mobile network coverage, so performance cannot be assured in areas without signal.

Operational Definition of Terms

Automated Queue Prioritization

The automatic ranking of patients in a service queue using a scoring algorithm based on defined criteria, without manual staff intervention.

Barangay Health Center (BHC)

A community-level public health facility operated by a local government unit that serves as the primary point of contact for residents seeking basic health services.

Digital Referral System

An electronic platform that facilitates the

transfer of patient information and referral requests between health facilities through standardized digital workflows.

Health Information System (HIS)

A digital platform designed to collect, manage, and transmit health data to support clinical and administrative decision-making across levels of care.

Intelligent Queue Priority Assignment

A module that computes a priority score per referred patient and dynamically updates the queue order in real time based on clinical urgency, demographic classification, and referral type.

Persons with Disabilities (PWDs)

Individuals with long-term physical, mental, intellectual, or sensory impairments. In this study, classified under Priority 2 in CareLink's queue prioritization algorithm.

Referral Completion Rate

The proportion of initiated patient referrals that result in the patient successfully presenting at and being attended to by the receiving health facility.

Rules-Based Algorithm	A computational process that applies predefined logical conditions and weighted criteria to produce a consistent, deterministic output.
Short Message Service (SMS) Notification	Automated text-based alerts delivered via mobile network. In this study, used for referral approval, 24-hour appointment reminders, and missed referral alerts.
System Usability Scale (SUS)	A standardized 10-item questionnaire that measures system usability from the end user's perspective, producing a score from 0 to 100.
Technology Acceptance Model (TAM)	A theoretical model predicting user adoption of information technology based on perceived usefulness and perceived ease of use.
Universal Health Care Act (RA 11223)	A 2019 Philippine law mandating LGUs to deliver accessible and quality health services to all Filipinos at the community level.
Web-Based System	A software application accessed through a

web browser over a network, requiring no local installation on the user's device.

Method

System Methodology

This chapter presents the development methodology of CareLink, detailing each phase undertaken in the design, construction, and evaluation of the system. It covers the essential aspects of the study, including requirements analysis, system design, software development, testing processes, implementation planning, and the results obtained throughout the development lifecycle.

Figure 3.

Research Methodology

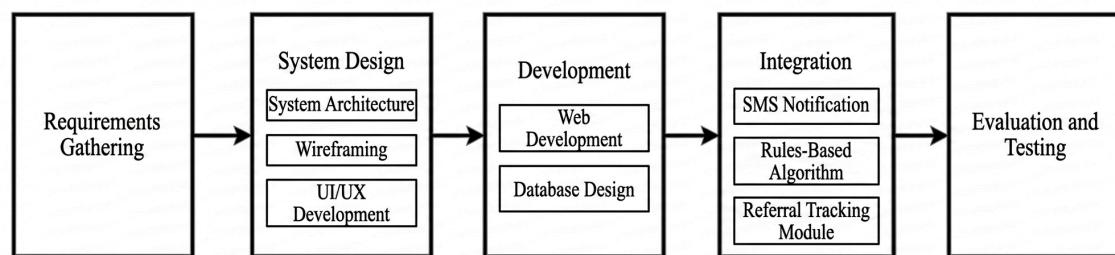


Figure 3 presents the process of CareLink development as stages that progress in a sequence, beginning with the requirements gathering process and proceeding to system design, development, and the integration of core services. An evaluation and testing process serves as the final step, determining the performance, usability, and efficiency of the system in addressing the identified gaps in the manual referral and queuing workflows of barangay and Koronadal City health centers.

Requirements Gathering

This phase concentrated on analyzing and documenting the needs and functions of CareLink's stakeholders: Barangay Health Center Staff, Koronadal City Health Center Staff, the Super Admin, and walk-ins/referred patients. The research team conducted structured interviews with health workers at selected Barangay Health Centers, the Koronadal City Health Center, and a provincial hospital outpatient department to understand how patients are queued, how referrals are prepared and tracked, and how severity is currently assessed. This informed an analysis of the existing manual process in Koronadal City, where patients queue physically without tracking, referrals are handled on paper, and completion cannot be verified. This phase shaped the functional and non-functional requirements of each module, including the digital referral form, severity classification, intelligent queue priority assignment, referral status tracking, automated SMS notification, and analytics dashboard.

System Design

The System Design phase transformed the gathered requirements into a comprehensive blueprint of CareLink by establishing the overall system architecture, defining how the web-based platform, database server, and integrated external services interact across the barangay and city health center interfaces. System architecture was structured following a three-tier web model composed of a presentation layer, an application logic layer, and a data layer, ensuring clear separation of interface, business rules, and data management responsibilities suited to low-bandwidth LGU health center environments. Wireframing was subsequently conducted to determine the structure and layout of interface elements across all user role dashboards before development commenced, while UI/UX development translated these wireframes into a clean,

accessible interface designed for health workers with varied digital literacy, directly applying the interface simplicity principles recommended by (Elepaño et al., 2025).

Development

In this phase, the construction of CareLink was carried out in accordance with the design specifications, covering the full web development of all system modules across the four user roles Super Admin, Barangay Health Center Staff, City Health Center Staff, and the patient-facing interface alongside the concurrent design of the relational database structure supporting core entities including Patient, Referral, User, HealthCenter, QueueEntry, SMSLog, and AuditLog. The Agile development methodology guided this phase, with module construction organized into iterative sprint cycles that allowed the research team to continuously test and refine individual components, including the rules-based queue priority scoring algorithm and the multi-state referral tracking workflow, rather than deferring all validation to the end of the development timeline.

Integration

This phase connected CareLink's core modules to the key external services and algorithmic components enabling its primary functions, including the SMS Notification gateway implementing a three-trigger workflow (referral approval alert, 24-hour appointment reminder, and missed referral alert) modeled on the system validated by (Presume et al., 2024). The Weighted Sum Model was integrated into the referral approval workflow to compute each patient's priority score from clinical urgency, demographic classification, and referral type, then insert them into a priority queue ordered by score and assign them to Priority 1 (Emergency), Priority 2 (Vulnerable

Group), or Priority 3 (Standard Referral) without manual intervention. The Referral Tracking Module established the multi-state workflow governing referral status from submission through completion, with integration testing conducted after each component connection.

Evaluation and Testing

The final phase assessed the functionality, usability, and overall performance of CareLink through a three-dimensional evaluation framework administered to Barangay Health Center Staff, City Health Center Staff, and the Super Admin following structured hands-on system use. System usability was measured using the ISO-recognized System Usability Scale (SUS), user satisfaction was assessed through a TAM-aligned post-use survey measuring Perceived Usefulness and Perceived Ease of Use grounded in the Technology Acceptance Model confirmed by (Tetik et al., 2024), and performance efficiency was evaluated through response time measurement of four key operations referral form submission, priority score computation, queue assignment, and SMS trigger latency to determine whether CareLink meets acceptable performance thresholds for deployment in the barangay and city health centers of Koronadal City.

Figure 4.

Agile Development (Amoros, 2025)

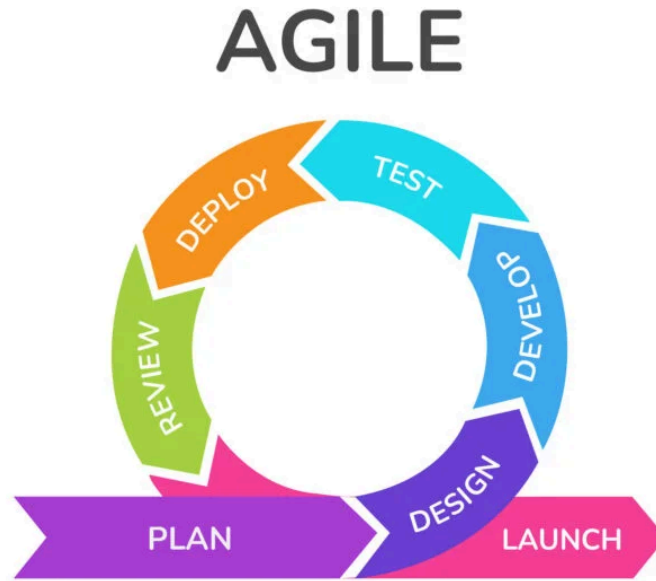


Figure 4 displays the Agile development life cycle used by the researchers in the development of CareLink. Agile is an iterative and incremental approach to software development that focuses on delivering working software using short time-bound development sprints. The reason why the Agile approach was chosen for developing CareLink was that this particular approach offered the research team flexibility and response to changes to meet changing needs of stakeholders, who were barangay and city health center personnel in relation to the development of CareLink in the course of four months.

Plan

During this phase, the overall project vision for CareLink was defined and decomposed into manageable sprint iterations, with system features ranked according to

functional priority and core modules such as patient registration, digital referral form, and queue priority assignment identified as the highest-priority deliverables. A product backlog was established to guide the sequence and scope of work across all subsequent development sprints, ensuring that the research team maintained a clear and organized roadmap throughout the entire development lifecycle of CareLink.

Design

The team collaboratively determined the system architecture and interface specifications to be addressed within each sprint, producing workable interface prototypes and data models for each module in scope rather than rigid long-term design documents. This iterative design approach allowed decisions to be continuously informed by early stakeholder feedback from barangay health workers and city health center staff, ensuring that the interface and system logic of CareLink remained aligned with the actual needs of its intended users at every stage of development.

Develop

The researchers implemented the designed features through active coding and module construction within each sprint cycle, maintaining continuous integration practices throughout development to preserve system functionality, stability, and progress transparency. Each sprint delivered at least one testable module, ensuring that development progress remained consistently aligned with the project timeline and that the rules-based queue priority scoring algorithm, multi-state referral tracking workflow, and automated SMS notification logic were each built and verified incrementally rather than all at once.

Test

Testing was conducted continuously alongside development rather than deferred to a final stage, with each module undergoing unit testing, integration testing, and user acceptance testing as it was completed to verify correctness before proceeding to the next sprint. This continuous testing approach ensured that critical CareLink components including the priority score computation logic, the referral status transition workflow, and the three-trigger SMS notification system were validated at the module level well before the full system integration and final evaluation phases.

Deploy

Upon successful testing of completed modules, the developed features were deployed to a staging environment for review by the research team and select stakeholders, allowing functional components of CareLink to be demonstrated and assessed for feedback well before the final system release. This iterative deployment approach ensured that any issues identified during staging reviews were addressed within the same or the immediately following sprint, maintaining a consistently stable and improving system build throughout the development period.

Review

At the close of each sprint, the research team presented completed features to stakeholders and advisers to assess alignment with the original system requirements and gather constructive improvement feedback, with lessons learned incorporated into the planning of subsequent sprints. This continuous review cycle ensured that CareLink's development remained responsive to the operational realities of Koronadal City's

barangay and city health centers, and that each iteration brought the system closer to a solution that genuinely addressed the identified gaps in the manual referral and queuing process.

Launch

This stage refers to the formal completion and release of CareLink for evaluation, with final preparations including database integrity verification, user account configuration for all respondent groups, and system performance validation to ensure reliable operation under evaluation conditions. The launch stage marked the culmination of the iterative Agile development process, confirming that CareLink was fully functional, stable, and ready for structured usability, satisfaction, and performance testing by its intended users in the barangay and city health center context of Koronadal City.

Requirements Analysis

In this section, the researchers identified the needs and expectations of the stakeholders by examining the existing manual referral process. The current workflow was analyzed to determine its challenges and limitations, ensuring that the proposed CareLink system addresses both the functional and non-functional requirements needed to improve referral coordination, queue management, and information exchange between Barangay and Koronadal City Health Centers.

Data Flow Diagram of the Current System

Figure 5.

Data Flow Diagram of the Current System

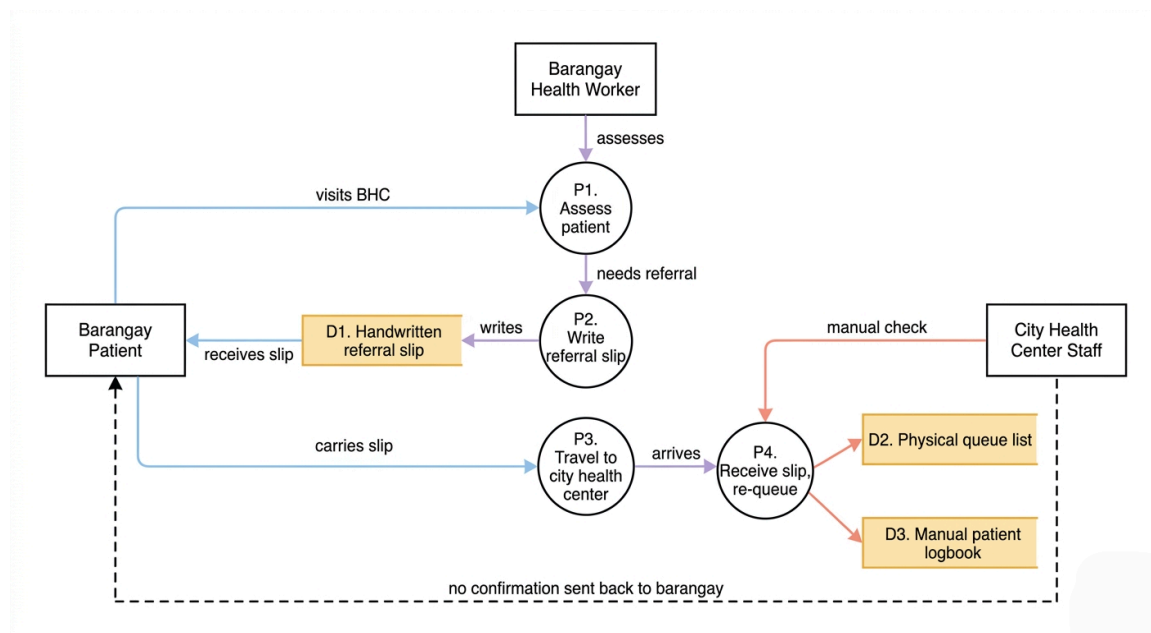


Figure 5 shows the information flow for the current referral system involving Barangay Health Centers and the Koronadal City Health Center. The flow starts with the visit of the patient to the Barangay Health Center where consultation and examination of the patient is conducted. In case the patient requires more attention from a specialist, the health worker at the Barangay Health Center manually completes the referral form with patient information. The patient will be in charge of delivering the referral to the City Health Center. The receipt of the referral at the City Health Center will be done manually by City Health Center Staff. The process will end after delivering the relevant health care service. No central computerized information system exists for referrals.

Use Case Diagram of the Current System

Figure 6.

Use Case Diagram of the Current System

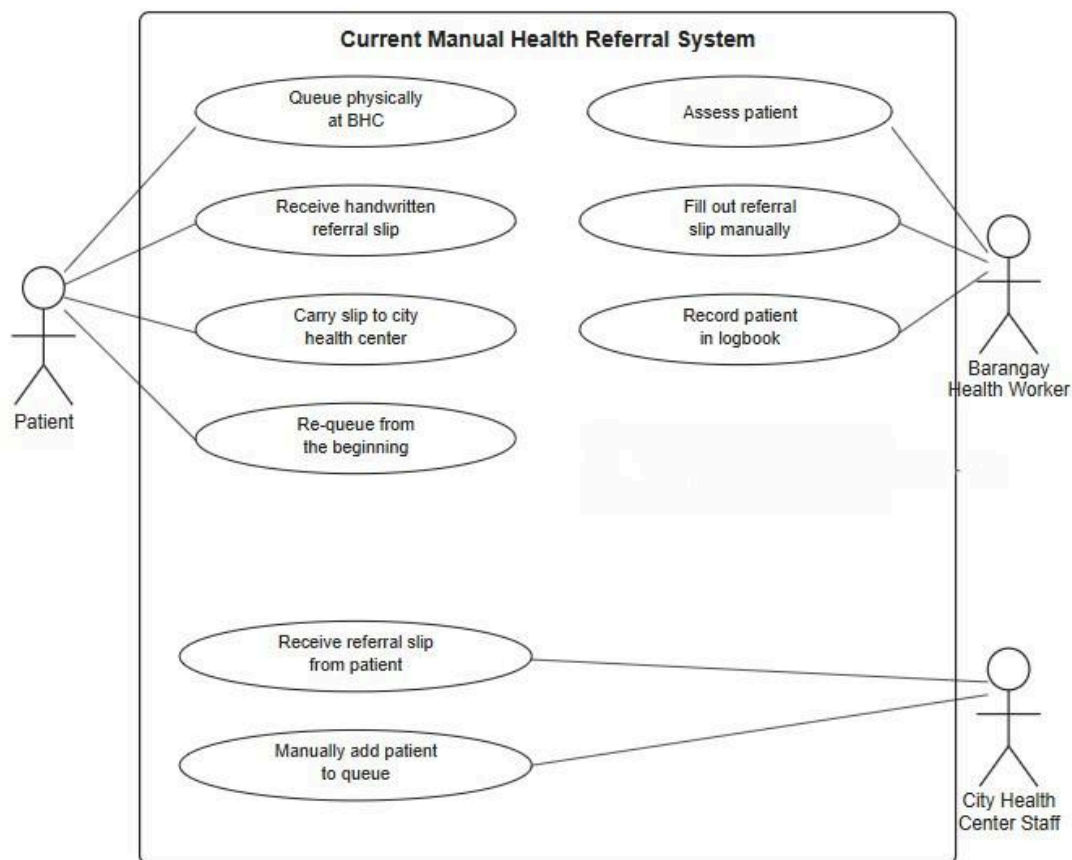


Figure 6 shows the flow of actions involved in the referral system between the Patient, Barangay Health Worker, and the City Health Center Personnel. The figure demonstrates that the referral involves the action of seeking consultation, getting referral slips, and then submitting the latter at the City Health Center for medical attention. What is evident from the flowchart is that the whole referral process is manual in terms of processing the referrals and managing the records and queue of patients.

Requirements Documentation

System Requirement for Hardware

Computer for CareLink Minimum System Requirements

- Intel Core i3 7th Gen
- 4GB RAM
- 512GB HDD
- Intel UHD Graphics 620
- Windows 7 OS or better

Recommended System Requirements

- Intel Core i5 7th Gen
- 8GB RAM
- 512GB SSD
- Intel UHD Graphics 620
- Windows 10 OS

Software

Visual Studio Code

Visual Studio Code is a free, open-source code editor developed by Microsoft that supports multiple programming languages and offers built-in features such as syntax highlighting, code completion, and version control integration. Its lightweight yet powerful environment makes it a suitable tool for developing CareLink: A Smart Health

Referral System with Automated Queue Prioritization for Barangay and Koronadal City Health Centers.

MySQL

MySQL is a relational database management system that stores and organizes the information required by applications. It uses Structured Query Language (SQL) to facilitate data extraction and manipulation.

Figma

Figma is a web-based software application that enables developers and designers to work together to create a UI design collaboratively and in real-time. The program comes with many functionalities such as wireframing, interactive prototyping, and component-based design. These capabilities can be applied when creating the user interface design of CareLink: A Smart Health Referral System with Automated Queue Prioritization for Barangay and Koronadal City Health Centers.

Vercel

Vercel is a cloud computing platform that enables the deployment and hosting of web applications. The key characteristics of the platform include automatic deployment, integration with GitHub, serverless functions, and CDN which delivers rapid access to web applications globally. Based on the above qualities, Vercel can be considered an appropriate hosting platform to deploy and publish CareLink: A Smart Health Referral System with Automated Queue Prioritization for Barangay and Koronadal City Health Centers.

Google Chrome

Google Chrome is a highly efficient and very secure web browser created by Google which is designed to accommodate current web standards. This browser has a number of advantages including developer tools, cross-platform usability, and frequent security updates that provide its users with maximum stability and performance optimization. This is why Google Chrome is the most suitable browser for CareLink: A Smart Health Referral System with Automated Queue Prioritization for Barangay and Koronadal City Health Centers.

Tailwind CSS

Tailwind CSS is a utility-based CSS framework utilized in designing user interface designs. Tailwind CSS is a framework that comes with predefined utility classes which help the developer to create their own unique design without necessarily creating new CSS code.

Node.js

Node.js is an open-source JavaScript platform which enables developers to run JavaScript code in the server-side.

User Requirements

Internet Connection. Stable internet connection is needed in order to be able to use the CareLink website and its functionalities including submitting referrals, checking patient data, and tracking the referral status.

User Account. Users of Barangay Health Center Staff, City Health Center Staff, and Super Admin roles need to have user accounts that will enable them to log into the website and perform their tasks effectively.

Web Browser. Users should use an updated web browser such as Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari to ensure compatibility and optimal system performance.

Computer or Mobile Device. The users of the health centers shall have either a desktop computer or laptop in case of the health center staff and a computer or mobile phone in the case of patients.

Mobile Number. The mobile number should be a working one to facilitate the sending of automated SMS notifications, which could be on referral acceptance or on missed referrals.

Patient Information. Accurate patient details, such as name, age, contact number, address, and health information, must be provided to ensure proper referral processing and record management.

Health Center Registration. Participating Barangay Health Centers and the Koronadal City Health Center must be registered in the system to enable referral submission, tracking, and coordination between facilities.

Device Permissions. Users accessing CareLink through devices may need to allow browser permissions such as internet access and notifications to ensure proper system functionality and timely updates.

Use Case Diagram of the Proposed System

Figure 7.

CareLink Super Administrator Use Case Diagram

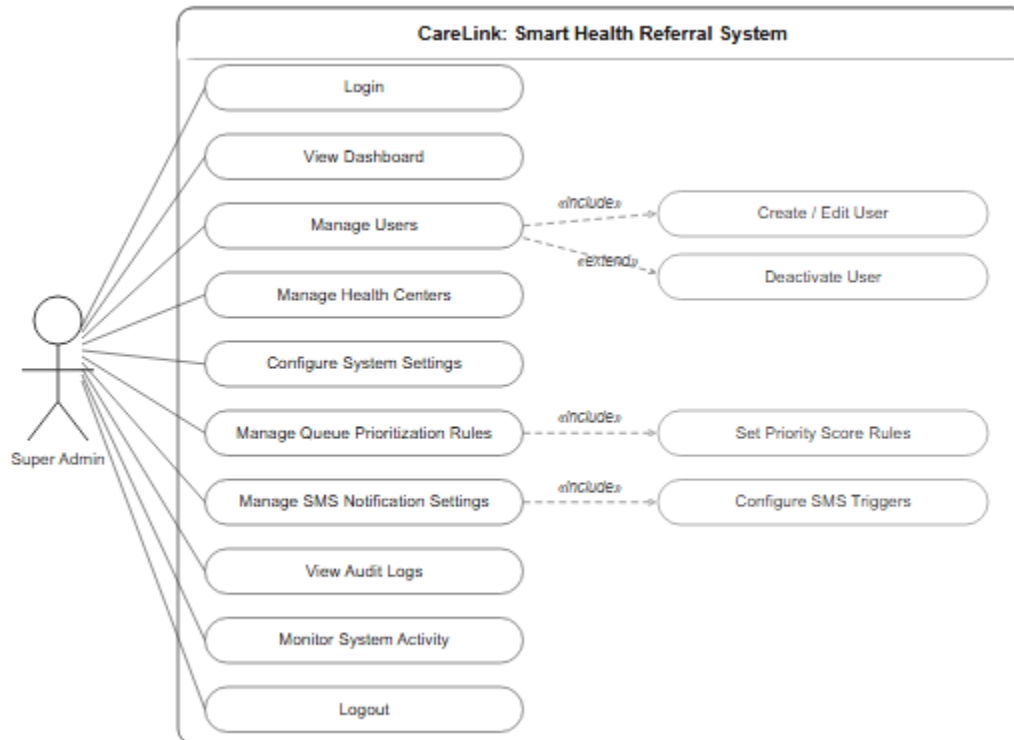


Figure 7 shows the use case diagram of the Super Administrator profile in the CareLink system, highlighting the management tasks performed by this user. As shown, the super admin is tasked with logging in, accessing the dashboard, and performing management operations on the users' account in the form of creation, editing, and deactivation of those accounts. Furthermore, the Super Admin is tasked with managing health centers' profiles in addition to configuring the general settings of the system and defining the prioritization criteria through setting the priority score threshold as described by the included use case. Lastly, the Super Admin manages SMS notification triggers and accesses the audit logs of the system before logging out.

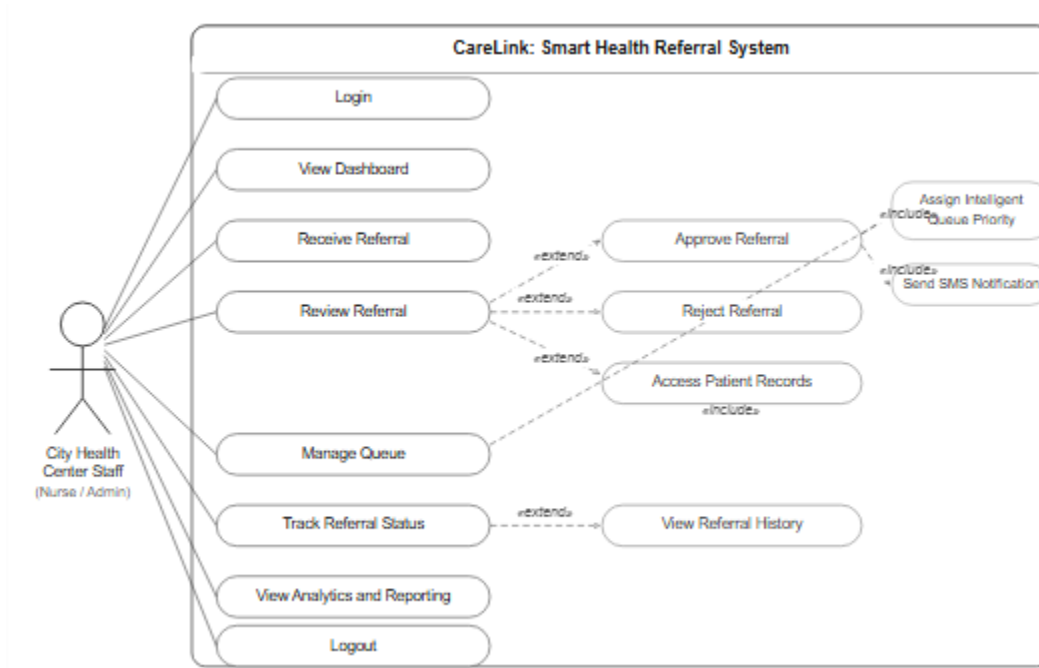
Figure 8.*CareLink City Health Center Use Case Diagram*

Figure 8 presents the use case diagram of the Barangay Health Center Staff role in CareLink, representing the nurses and midwives who serve as the primary referring actors in the system. This role is responsible for logging in, viewing the dashboard, and registering patients by entering patient details through an included use case. The Barangay Health Center Staff manages patient records, with the option to update health records as an extended action. They submit digital referral forms by encoding the referral reason as a required included step, replacing the manual handwritten referral slip process. Staff can also track the real-time status of submitted referrals, view referral history for monitoring and documentation purposes, and log out of the system. This role centralizes the referral initiation workflow within a single digital interface.

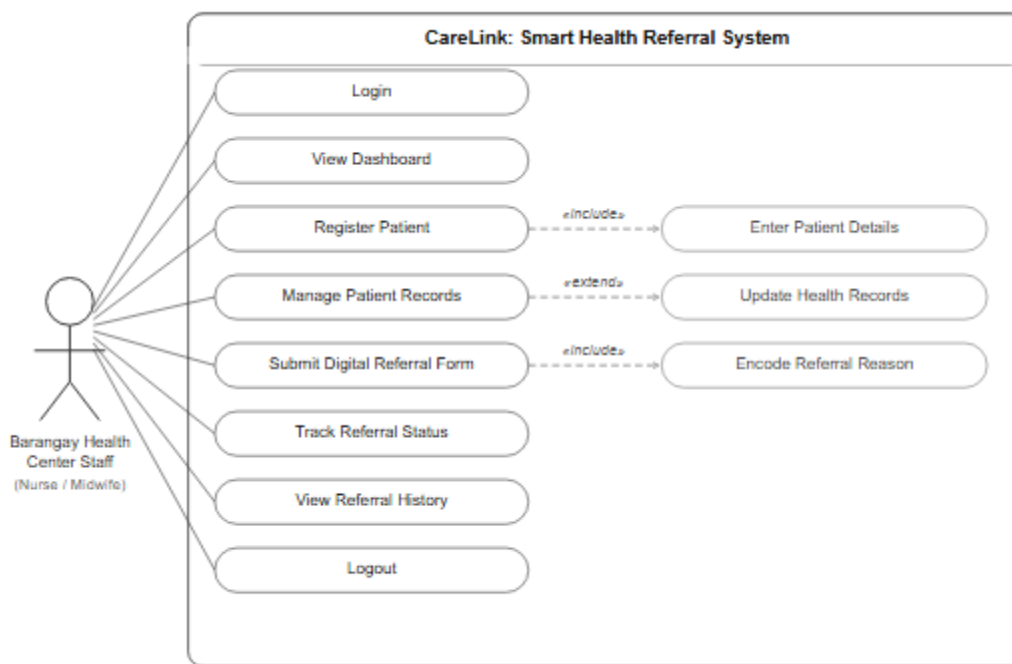
Figure 9.*CareLink Barangay Health Center Use Case Diagram*

Figure 9 presents the use case diagram of the City Health Center Staff role in CareLink, representing the nurses and administrators who receive and process incoming referrals. This role logs in and views the dashboard before receiving and reviewing referrals. Upon review, staff may approve a referral, which automatically triggers two included actions: the assignment of an intelligent queue priority score and the sending of an SMS notification to the patient. Alternatively, staff may reject a referral as an extended action. Staff can also access patient records as part of referral review. The Manage Queue function allows staff to oversee the prioritized patient queue, while the Track Referral Status function extends to viewing referral history. The role also includes access to the Analytics and Reporting module for monitoring referral volumes, queue performance, and service efficiency before logging out.

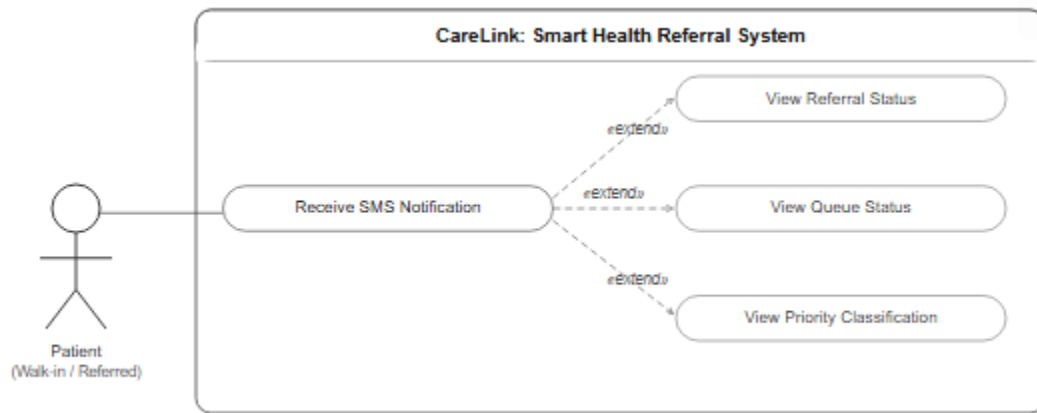
Figure 10.*CareLink Patient Use Case Diagram*

Figure 10 shows the use case diagram for the Patient role within CareLink, including both walk-in and referred patients who engage with the application. The main use case involving the Patient role in CareLink is receipt of SMS notifications, which form the core use case from which there are three extended use cases that occur. After receiving an SMS notification, the patient has the option of viewing his or her referral status to keep track of how far the referral submission has gone, viewing queue status to determine what the individual's position in the queue is, and viewing priority status to find out the priority assigned to them according to the rules-based algorithm.

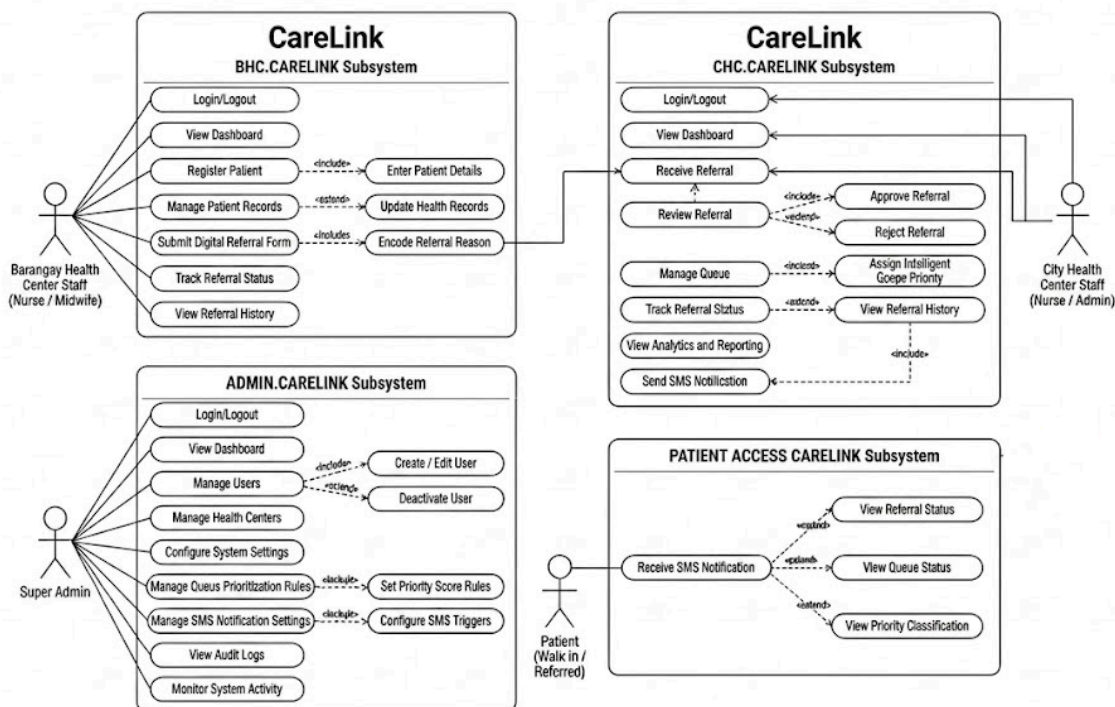
Figure 11.*CareLink Smart Health Referral System Use Case Diagram*

Figure 11 shows the combined use case diagram of the CareLink Smart Health Referral System, incorporating the interactions between all key actors into one. The diagram shows how Barangay Health Center Staff conducts activities related to patient registration, patient recordkeeping, referral submission, referral tracking, and referral history; how the City Health Center Staff performs activities such as receiving referrals, reviewing referrals, approving or rejecting referrals, patient queuing, referring status tracking, generating analytics/reports, and sending SMS notifications; how the Super Admin performs activities such as user management, health center management, configuration of the application, queue priority rules, SMS notification rules, audit logging, and system activity monitoring; and how the Patient receives SMS notifications

and views referral status, queuing status, and priority level. This combined diagram demonstrates the full process of CareLink, which refers to digital referral management in the Philippines that involves transmitting referrals to the City Health Center, processing them through queue management and SMS notifications systems, and managing them using administrative features.

Activity Diagram

Figure 12.

Activity Diagram of CareLink

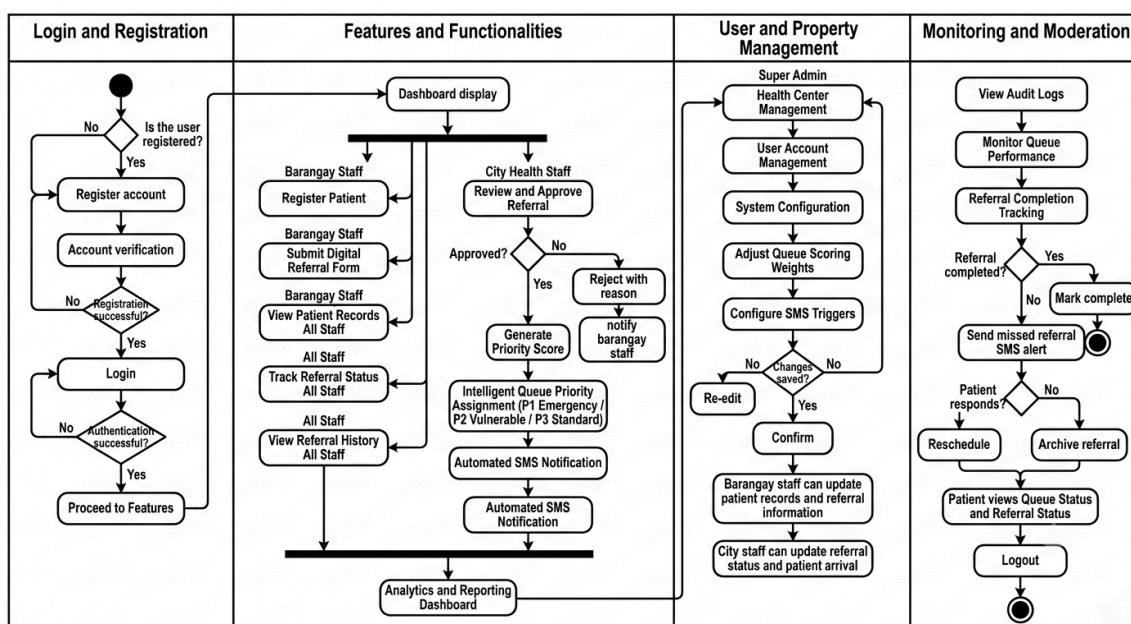


Figure 12 presents the activity diagram of the CareLink Smart Health Referral System, illustrating the overall workflow from user authentication to referral processing, queue prioritization, administrative management, and patient monitoring. The process starts with user registration and login, where there will be user verification and authentication prior to accessing the system features. Barangay Health Center Staff may

register patients, submit referral forms, see patient data, monitor referral statuses, and check referral history. On the other hand, City Health Center Staff evaluates received referrals and approves/rejects them; approval of referrals leads to the prioritization and scoring of queues depending on whether it falls under Priority 1 (Emergency), Priority 2 (Vulnerable), or Priority 3 (Standard). Additionally, there will also be automatic messages sent to patients via SMS upon receiving approval of referrals. The Super Admin is tasked with managing the health centers, users, system, scoring queue weights, and SMS triggers in order to ensure that there is proper system management and operations. In terms of monitoring and moderation, stakeholders should expect to see audits logs, queue monitoring, referral monitoring, sending missed referral SMS notifications, and archiving unhandled referrals. During the referral process, patients would be able to see their referral and queue status via SMS notifications.

Modules of the System

Super Admin Modules

Login Module

This module allows only authorized system administrators to access the web-based CareLink administration interface. It enforces secure authentication to ensure that only designated personnel can execute system-level management functions and access sensitive health center configuration data.

Dashboard Module

This module provides the system administrator with a centralized overview of CareLink's operational status, displaying key system metrics, active user counts, health center activity summaries, and quick-access links to all administrative management functions.

User Management Module

The module provides the means for the system administrator to add, modify, disable, and control the user accounts, regardless of their function within the system, from Barangay Health Center Staff to City Health Center Staff, thereby allowing only those who are authorized to access the system.

Health Center Management Module

This is the module by which the system administrator can register and configure the profile of the involved barangay health centers and the Koronadal City Health Center within the system.

System Configuration Module

This module enables the system administrator to define and adjust system-wide settings, including queue prioritization scoring weights, SMS notification triggers, and referral workflow parameters, allowing CareLink to be adapted to evolving operational requirements without modifying the codebase.

Audit Log Module

This module automatically records all significant system activities and user actions across all roles, providing the system administrator with a traceable history of transactions, configuration changes, and access events for accountability and security monitoring purposes.

Barangay Health Center Staff Modules

Login Module

The log-in module serves the purpose of authenticating staff members from the barangay health center through security measures that allow only those nurses and midwives registered in their respective barangay health centers to gain access to the system.

Dashboard Module

This module presents Barangay Health Center Staff with a real-time summary of their facility's referral activity, including pending referrals, recently submitted referral statuses, and patient registration counts, enabling staff to monitor daily operations at a glance.

Patient Registration Module

This module allows Barangay Health Center Staff to create and maintain digital patient profiles within CareLink, recording essential demographic and medical background information that serves as the foundation for referral form generation and patient records management.

Severity Classification Module

This module allows the Barangay Health Center nurse or midwife to record an initial severity assessment based on the patient's presenting condition during consultation. The severity level is submitted with the digital referral form and serves as a weighted input to the City Health Center's priority scoring computation, carrying front-line clinical judgment into the automated prioritization process.

Digital Referral Form Module

This module provides the capability for Barangay Health Center Staff to encode and submit their structured digital referral forms to the Koronadal City Health Center using a structured electronic form with information on the patient's condition, severity classification, and reason for referral.

Referral Status Tracking Module

This module helps the Barangay Health Center Staff to track the current status of all their submitted referrals in real time whether the referral is still under consideration, rejected, approved, or completed by the receiving city health center.

Patient Records Module

This module offers the opportunity for Barangay Health Center Staff to access the health record of their referred patients to be able to check and modify patient data and referral history.

Referral History Module

This module maintains a complete referral history of referrals sent by the barangay health center.

Koronadal City Health Center Staff Modules

Login Module

This module provides secure authentication for Koronadal City Health Center Staff, ensuring that only authorized nurses and administrative personnel assigned to the city health center can access the referral receiving and queue management functions of the system.

Dashboard Module

This module gives City Health Center Staff a centralized real-time view of incoming referrals, current queue status, priority assignments, and daily referral activity, allowing staff to immediately identify pending actions and monitor patient flow at the city health center level.

Referral Receiving Module

This module consolidates all incoming digital referrals submitted by participating barangay health centers into a structured receiving queue, replacing informal and phone-based referral coordination with a standardized incoming referral management interface for city health center staff.

Referral Review and Approval Module

This module enables City Health Center Staff to review the details of each incoming referral, assess completeness and clinical information, and either approve or reject the referral with a documented reason, initiating the automated queue assignment and SMS notification workflows upon approval.

Intelligent Queue Priority Assignment Module

This module computes each approved patient's priority score using a Weighted Sum Model, combining clinical urgency, demographic classification, and referral type into a single weighted score. Patients are ordered in a priority queue by descending score, with timestamp as a tiebreaker, and automatically assigned Priority 1 (Emergency), Priority 2 (Vulnerable Group), or Priority 3 (Standard Referral), reflected dynamically in the queue without manual staff action.

Patient Records Module

This module allows access of City Health Center Staff to the digital health records of the referred patients including their registrations and previous referrals along with the prior barangay assessment.

Referral History and Tracking Module

This module helps maintain the full history of the referrals made to the city health center, enabling the staff to analyze the results of the referrals and also view past referral documentation when needed.

Analytics and Reporting Dashboard Module

This module provides visualizations and data reports on various analytics parameters related to the city health center like referral counts, queue statistics, priority statistics, referral result percentage, and SMS notifications statistics. In addition, the module provides the following advanced analytics features:

Peak Hour Analysis. Identifies the specific hours and days with the highest volume of referrals and walk-in patients, allowing health center administrators to anticipate demand and plan staff allocation accordingly.

Average Service Time per Transaction. Computes the average duration between a patient's queue entry and the completion of their consultation or referral, per priority tier, allowing administrators to identify bottlenecks in service delivery.

Staff Performance Monitoring. Tracks the number of referrals reviewed, approved, or processed per staff member over a given period, supporting workload distribution and accountability.

Queue Abandonment Rate. Measures the proportion of patients who were assigned a queue number but did not complete their consultation, helping administrators identify patterns of patient drop-off and their possible causes.

Patient Satisfaction Trends. Aggregates post-visit patient feedback ratings over time to visualize whether satisfaction with referral and queueing experience is improving, declining, or stable.

Predictive Queue Volume Forecasting. Uses historical referral and queue data to project expected patient volume for upcoming days or hours, allowing health centers to anticipate periods of high demand in advance.

Patient Interface Modules

Queue Status Viewing Module

This module allows walk-in and referred patients to view their current queue number, priority classification, and estimated position in the city health center queue in real time, eliminating the need to physically wait at the facility without any information about their service status.

Referral Status Viewing Module

In this module, referred individuals can view the status of the referral form that has been sent to the city's health center, which helps eliminate any confusion and allows the individual time to prepare before visiting the health center as per his schedule.

Design of the System

System Architecture

Figure 13.

System Architecture

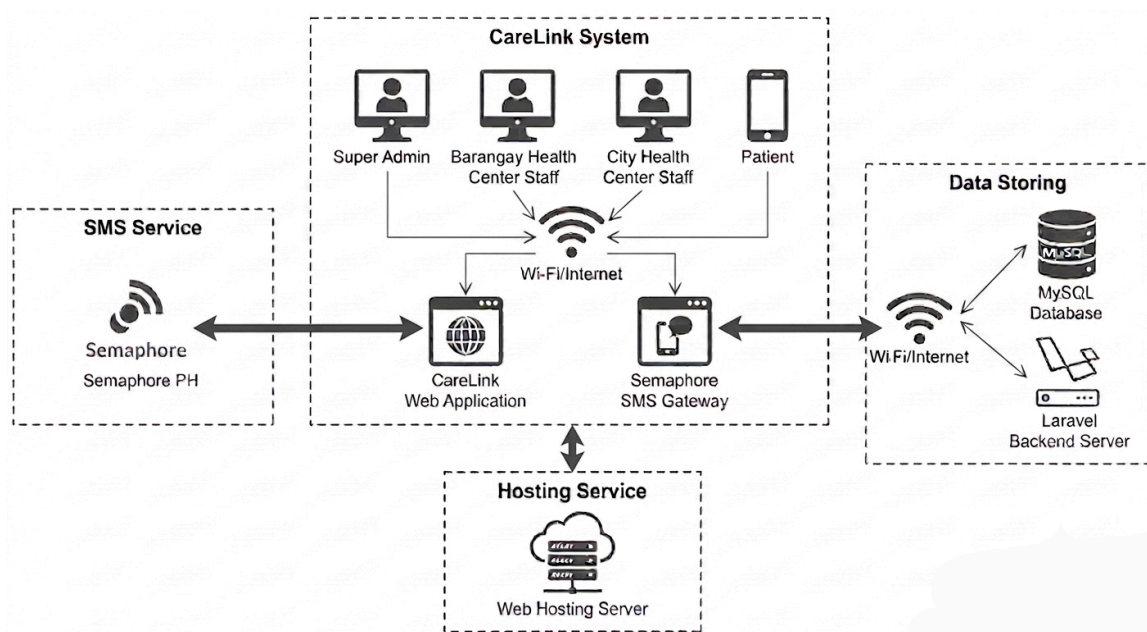


Figure 13 shows the system architecture of CareLink that depicts how the system is used and how user interaction is integrated with the application services, data management component, and external communication services. As depicted in Figure 13, the system is designed for four major types of users: the Super Admin, Barangay Health Center Staff, City Health Center Staff, and Patients who will use the system either via desktop or mobile device connected to the internet. The user inputs go through the CareLink Web Application, which acts as the interface for the whole system and runs from the web hosting server. The application works by interacting with the Laravel Backend Server where business logic, referrals, queue priorities, user accounts, and other functionalities of the system are done. At the same time, the data of the system are stored using the MySQL Database. For patient notification, the system uses the Semaphore SMS Gateway, which then connects with the external Semaphore PH SMS Service to send automatic notifications such as approval of referrals, queue priority, appointment reminder, and missed referral messages.

Deployment Diagram

Figure 14.

Deployment Diagram

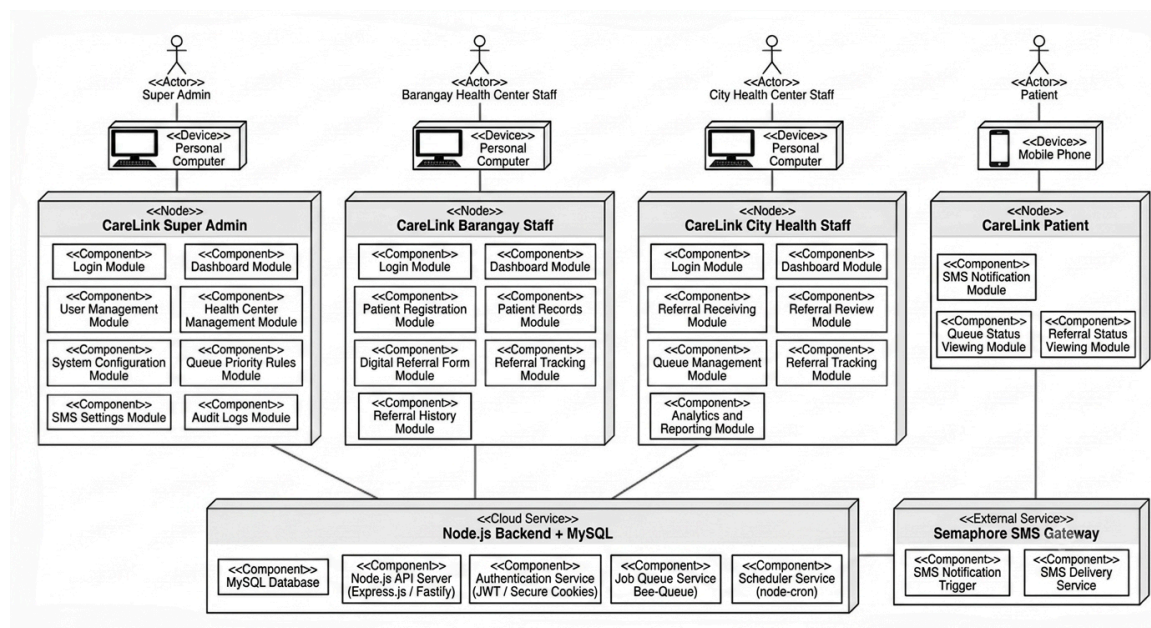


Figure 14 below shows the deployment diagram for CareLink where it is possible to see the physical arrangement of all the different components within the system and how they interact with each other. The Super Admin, Barangay Health Center staff, and City Health Center staff access their respective modules through personal computers, while patients get information about referrals through mobile phones. All these different UIs connect to the central Laravel Backend and MySQL Database that handle all aspects of security, data processing, and data storage in the application. The Laravel backend is also linked to the Semaphore SMS Gateway that manages the process of sending SMS messages to patients.

Entity Relationship Diagram

Class Diagram

Data Schema of the System

Data Dictionaries

Results and Discussions

Development and Testing

System Usability Scale (SUS)

Description of the System

Implementation Plan

Recommendations

As further extensions of the study, the following are recommended.

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APPENDICES

Appendix A Screenshots of the Testing of the Prototype

Appendix B Official Testing Results (optional)

Appendix C Evaluation Responses (optional)

Appendix D Pictures showcasing the actual design/development of the prototype

Appendix E Forms

(this must be inserted on top of FORM 01)

ORDER OF APPENDICES

- A. Screenshots of the testing of the prototype
- B. Evaluation Forms/Questionnaires (for research with Evaluation)
- C. Survey Responses
- D. Pictures showcasing the actual design/development of the prototype/conduct of the survey
- E. Sample Reports from the System (if any)
- F. Insert other Appendices here if necessary
- G. Form 01 – Thesis Title Approval Sheet
- H. Form 02 – Invitation to Thesis Adviser
- I. Form 03 – Invitation to Thesis Panelist (3)
- J. Form 14 - Thesis Revision Chart (Proposal Defense)
- K. Form 14 - Thesis Revision Chart (Pre-Final Defense)
- L. Form 14 - Thesis Revision Chart (For Groups With Additional Revisions During Final Defense)
- M. Form 06 - Consultation Record with Thesis Adviser
- N. Form 08 - Endorsement For Thesis Proposal or Final Defense
- O. Form 20 – Endorsement For The Reproduction Of The Thesis Paper
- P. Form 22 - Certificate Of Authentic Authorship
- Q. Technical Reviewer (c/o Ma'am Sueno)
- R. Source codes of the Program - necessary modules only (for projects with the program only)
- S. Pictures of Proposal Defense, Pre-Final Defense, and Final Defense

T. One-page Curriculum Vitae per team member (Design a CV using Canva)

SAMPLE CV



HAJAH T. SUENO

PROFILE

Recently earned a doctorate degree in Information Technology with an emphasis in data mining, machine learning, and information management science. Dedicated, hard-working and passionate about big data. Published and presented scholarly research papers in international Scopus indexed journals and conferences.

CONTACT

- +639772534415
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SKILLS

- Knowledge of standard research concepts in IT
- LaTeX Editing
- Mendeley
- Web Development and Hosting
- Joomla
- Adobe Photoshop
- C++ Programming
- MS Office Suite

AWARD

- **Excellent Paper Award**
World Research Forum for Advances in Science and Engineering, Jakarta, Indonesia. February 27-28, 2020

MEMBERSHIPS

- **Research Committee Member**
The International Institute of Engineers and Researchers (TheIRES)
- **Member (2015 - 2017)**
Association for Computing Machinery (ACM)
- **Regional Treasurer (2009 - 2017)**
Philippine Society of Information Technology Educators (PSITE)

WORK EXPERIENCE

Professor

Notre Dame of Marbel University, | November 2004 - Present

- Conduct instruction, consultation and other student-related activities in and out of the classroom in the areas of programming, Web Development, Data Structures and Algorithms, Software Engineering, and Capstone Projects.
- Advise student organization in the crafting and implementation of projects and activities.
- Conduct and deliver IT-related curriculum development, workshops and seminars, and community extensions in and outside the university.

EDUCATION

Doctor in Information Technology | June 2017 - March 2020

Technological Institute of the Philippines | Quezon City, PH

Dissertation: Enhanced Bayesian Vectorization Technique in Transforming Text Document to Numeric Features for Multi-class Text Classification using Support Vector Machine

Master of Science in Information Technology | June 2002 - March 2004

Notre Dame of Marbel University | Koronadal City, South Cotabato, PH

Bachelor of Science in Computer Science | June 1997 - March 2002

Notre Dame of Marbel University | Koronadal City, South Cotabato, PH

PUBLICATIONS

AUTHOR: Converting Text to Numerical Representations using Modified Bayesian Vectorization Technique for Multi-class Classification. International Journal of Advanced Trends in Computer Science and Engineering (IJATCSE). Vol 9, No.4., <https://doi.org/10.30534/ijatcse/2020/211942020>
 • **Presentation:** Bangkok, Thailand, December 21-22, 2019

AUTHOR: Dimensionality Reduction for Classification of Filipino Text Documents based on Improved Bayesian Vectorization Technique. International Journal of Advanced Trends in Computer Science and Engineering (IJATCSE). Vol 9, No.5. <https://doi.org/10.30534/ijatcse/2020/87952020>
 • **Presentation:** Jakarta, Indonesia. February 27-28, 2020

AUTHOR: Multi-class Document Classification using Support Vector Machine (SVM) Based on Improved Naive Bayes Vectorization Technique. International Journal of Advanced Trends in Computer Science and Engineering (IJATCSE). Vol 9, No.3. <https://doi.org/10.30534/ijatcse/2020/216932020>

CO-AUTHOR: An Empirical Analysis of Political Party Dominance in the Philippines Using Markov Chain and GIS Mapping. 14th National Conference of Information Technology Education (NCITE), Dipolog City, Philippines. October 2016. <https://bit.ly/3chgnFF>

